

SymPy Bug Card

Bug 43: Collapsed Range for $\cos(x)/\cot(x)$

Task. Compute the real range of the quotient

$$f(x) = \frac{\cos x}{\cot x}.$$

SymPy's answer.

$$\text{function.range}(f, x, \mathbb{R}) = \{0\}.$$

Correct answer.

$$\text{range}(f) = (-1, 1).$$

Explanation. At the defined point $x = \pi/3$, the quotient evaluates to $\sqrt{3}/2$, so the range cannot be $\{0\}$. The cancellation $\cos x / \cot x = \sin x$ is valid only on the original quotient's domain; points $x = k\pi/2$ are excluded, including the points where $\sin x = \pm 1$.

Likely root cause. The diagnosis identifies a trig rewrite in `sympy/calculus/singularities.py` that rewrites away the denominator condition $\cot(x) = 0$; then `sympy/calculus/util.py` treats the period as a closed interval and constructs the range from endpoint and critical-value sampling. This is a new failure mode with no public precedent.

Artifact (GitHub). [bug-report/bug-043-collapsed-range-for-cos-x-cot-x/](#)